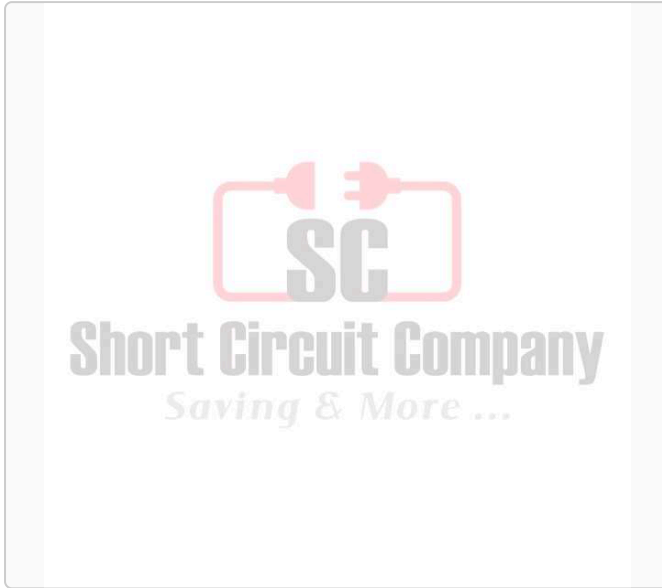




ABSTRACT

Short circuit Highbay for industrial use.

High performance of 98% of led efficiency within more than 50,000 operating hours.

Highbay is protected against the electricity instability with over voltage, current, and temperature protection.

Highbay consists of Philips chips and short-circuit drivers with warranty of 5 years.

SC HighBay Lighting 100W

High lux, High performance, High protection with long life-span for industrial use.

Short Circuit Company



Table of Contents

1 General product information	
1.1 General Specifications	
1.2 Environmental Compliance	
1.3 Compatibilities	
1.4 Product Main Parts	
2 Body and Heat-Sink	
2.1 Heat-Sink Operation Theory	
2.2 Body and Heat-Sink Description	
3 LED Chips	
3.1 Chip Performance Characteristics	
3.2 Electrical and Thermal Characteristics	
3.3 Absolute Maximum Ratings	
3.4 Characteristics Curves	
3.4.1 Light Output Characteristics	
3.4.2 Forward Current Characteristics	
3.4.3 Polar Curve	
4 Driver	
4.1 General Description	
4.2 Features	
4.3 Protections	
4.4 Module Specification	
4.4.1 AC Input Specification	
4.4.2 DC Output Characteristics	
4.4.3 Protection Specifications	
4.4.4 Mechanical Specification	



1. Product General Information

1.1 General Specifications:

Table1: Product General Specifications

Item	Specification	Unit
LED Chip	Philips 3030	
Driver	S.C	
LED efficiency	145	lm/W
Wattage	100	W
Total Luminous	14,500	lm
Beam angle	120Degree	°
Color temperature	6500K	K
CRI	>90	
Power factor	0.95	
Input voltage	AC ~ 8 5-300 V	V
Frequency	50-60Hz	Hz
Operating Temperature	-40 ~ +50	°C
Humidity	Up to 90 %	
IP Rating	65	
Driver Protection	OVP, OCP, OTP	
Life time	50,000Hour	Hours
Warranty	5Year	Years



1.2 Environmental Compliance:

Short Circuit is committed to providing environmentally friendly products to the solid-state lighting market. Highbay LED is compliant to the general environmental duty (GED). Short Circuit will not intentionally add the following restricted materials to its products: lead, mercury, cadmium, hexavalent chromium, polybrominated biphenyls (PBB) or polybrominated diphenyl ethers (PBDE).

1.3 Compatibilities

RoHS ISO CE UL LM79 LM80

1.4 Product Main Parts:

Highbay LED is mainly consisted of three parts described as following:

Body and heat sink which is made of Aluminum alloy to dissipate the internal temperature of Chips and Drivers.

Philips 3030 LED Chips with efficiency of 145Lm/W.

Short Circuit Driver which is responsible for supplying the chip with required voltage and current. Also, the driver is protected against over voltage and over current occurrence.

2. Body and Heat-Sink

2.1 Heat-Sink Operation Theory

Due to electronic circuits nature, there is over internal heat generated during the operation period. That heat needs to be dissipated to protect driver components from being damaged.

According to Heat Transfer Theory, materials with high thermal conductivity are used to transfer heat from the hot region to the cold one, space in our case, so a heat-sink made from Aluminum alloy is a suitable choice for this mission.

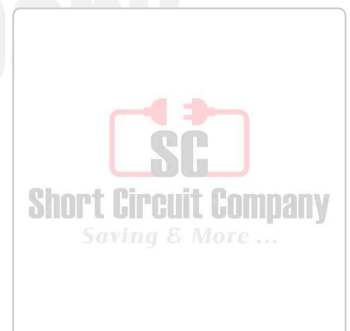


Figure1: Body and Heat-Sink



2.2 Body and Heat-Sink Description:

Table2: Body and Heat-Sink Description

Item	Description	
Body	Anti-corrosion aluminum alloy heat sink (electrostatic painted)	
Aluminum Purity	96-98 %	
Number Of Fins	48	
Thickness Of Fin	Inner Outer	1.5mm
Lenses	UV Poly-carbonate heat resistant lenses	
Beam Angle	120Degree°	
Gasket	silicone rubber insulation	
Glands	Waterproof cable glands (silicone insulated)	
IP Rating	IP65	
Weight	3.7kg kg	
Wattage	100 W	

3. LED Chips

3.1 Chip Performance Characteristics

Table3: Chip Performance Characteristics

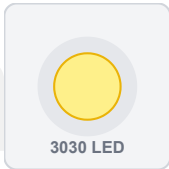


Figure2: Philips Lumiled 3030

Product	Nominal CCT	Minimum CRI	Typical Luminous Efficacy (lm/W)	Luminous Flux (lm)		Part Number
				Minimum	Typical	
LUXEON 3030 2D (Square LES)	6500K	90	144	94	104	L130-6590003000X21



3.2 Electrical and Thermal Characteristics

Table4: Chip Electrical and Thermal Characteristics

Part Number	Forward Voltage (Vf)			Typical Temperature Coefficient of Forward Voltage [2] (mV/°C)
	Minimum	Typical	Maximum	
L130-6590003000X21	5.8	6	6.6	-2 to -4

3.3 Absolute Maximum Ratings

Table5: Chip Absolute Maximum Ratings

Parameter	Maximum Performance
DC Forward Current	240mA
Peak Pulsed Forward Current	300mA
LED Junction Temperature (DC & Pulse)	125°C
Operating Case Temperature	-40°C to 105°C
LED Storage Temperature	-40°C to 105°C

3.4 Characteristics Curves

3.4.1 Light Output Characteristics

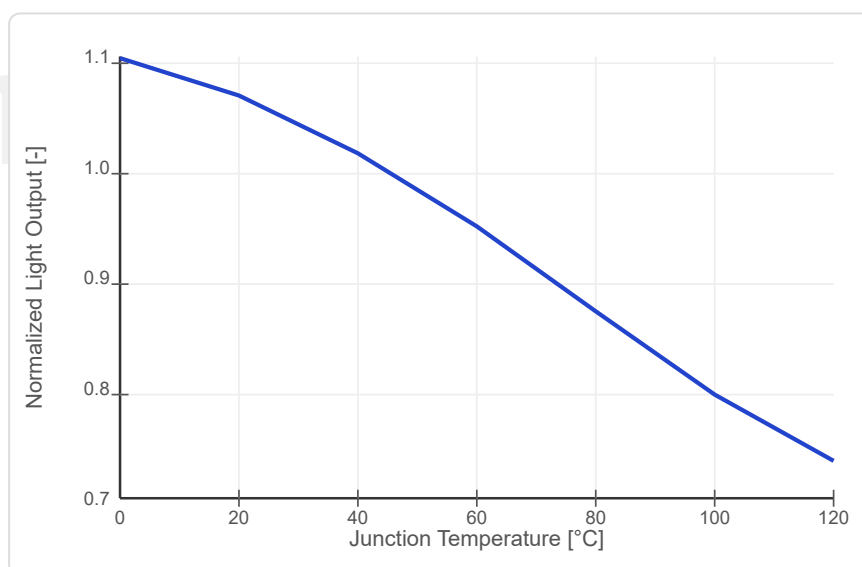


Figure3.a: Chip Light Output Characteristics

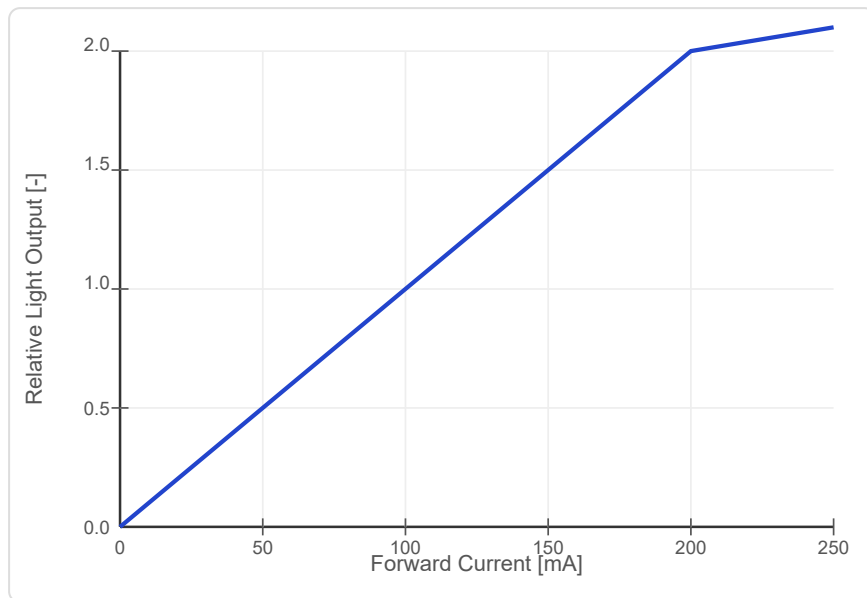


Figure3.b: Chip Light Output Characteristics

3.4.2 Forward Current Characteristics

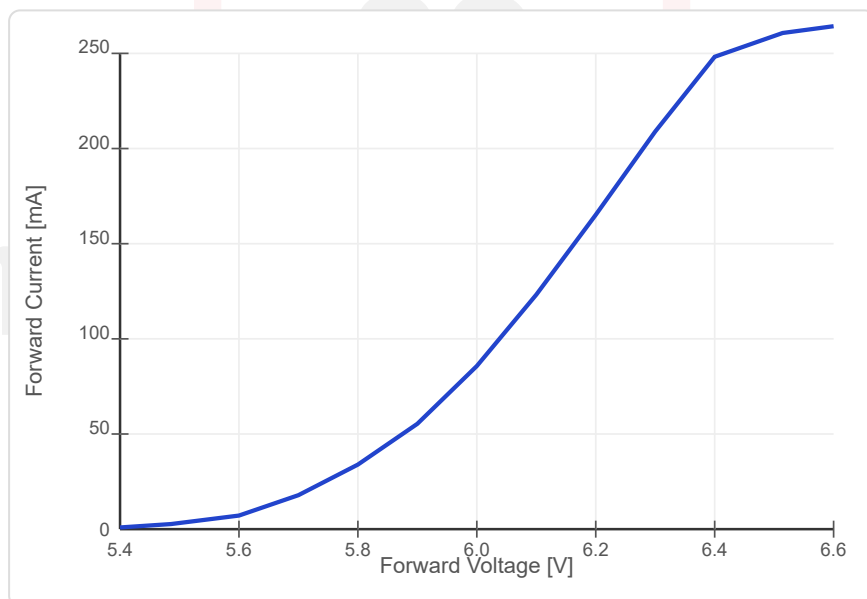


Figure4: Chip Forward Current Characteristics



3.4.3 Polar Curve

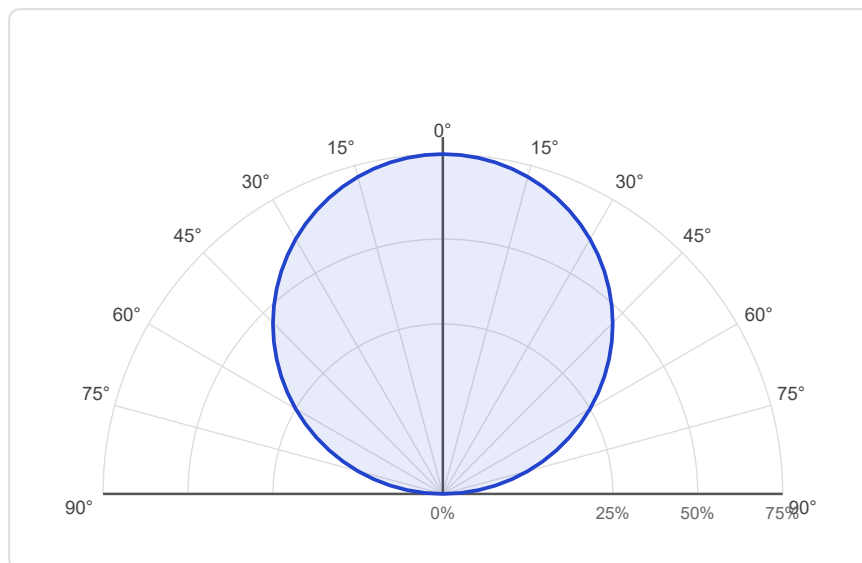


Figure5: Chip Polar Curve

4. Driver

Model number	SC-LDF-35V1450mA
Type	Universal isolated LED driver

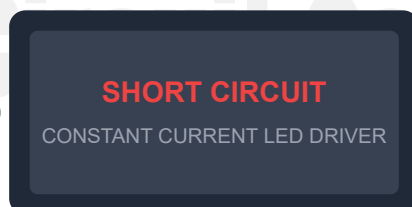


Figure6: Short Circuit Driver



4.1 General Description

LDF series is an offline LED lighting controller with high power factor, low THD and high constant current (CC) precision. It can achieve low system cost for an isolated lighting application by primary side control in a single stage converter. It significantly simplifies the LED lighting system design by eliminating the secondary side feedback components and the opto-coupler.

The proprietary CC control is used, and the system can achieve high power factor with constant on-time control. Quasi-resonant (QR) operation and clamping frequency improve the system efficiency.

The advanced start-up technology is used to meet the start-up time requirement (<0.5s). The constant output current is compensated for tolerance of transformer inductance variation. Besides, the line compensation and load compensation are built in for high precisely constant output current control.

This series offers comprehensive protection coverage with auto-recovery features including programmable VIN foldback, programmable VIN OVP, programmable thermal foldback, LED open loop protection, LED short circuit protection, cycle-by-cycle current limiting, built-in leading-edge blanking, VDD under voltage lockout (UVLO).

4.2 Features

- ✓ **Primary-side control with single stage PFC LED driver for low system cost,**
- ✓ **High PF (>0.95),**
- ✓ **Low THD (<10%),**
- ✓ **High efficiency (>85%),**
- ✓ **Life time up to 50,000 hours**
- ✓ **Pre-programmed VIN foldback,**
- ✓ **Pre-programmed VIN OVP,**
- ✓ **Pre-programmed thermal foldback,**
- ✓ **Pre-programmed CC regulation,**
- ✓ **High precision constant current,**
- ✓ **Fast start-up time (<0.5s),**
- ✓ **Built-in line/load compensation,**
- ✓ **No visible flicker, and audible noise free**

4.3 Protection

- ✓ **LED short circuit protection,**
- ✓ **LED open circuit protection,**
- ✓ **VDD over/under voltage protection,**
- ✓ **Lightning surge voltage up to 10 kV,**
- ✓ **AC input overvoltage protection,**
- ✓ **AC input overcurrent protection.**

The LED driver recovers automatically and back to normal operation after all fault causes are removed.



4.4 Module Specification

This section presents the specification of the LED driver, normal operating range, and the driver performance.

4.4.1 AC Input Specification

Table6: AC input specification

LED driver max. apparent power	56W at full load 95Vac<Vin<275Vac
AC input voltage rating	30Vac ~ 300Vac
AC input voltage range for normal operation	90Vac ~ 275Vac
AC input voltage range for foldback operation	30Vac ~ 90Vac
AC input frequency range	47Hz ~ 63Hz
Power factor	>0.95 at 220Vac full load
AC input current THD	<10% at 220Vac full load
AC max. input current	<0.8A at 95Vac full load
Efficiency	>85% at 220Vac full load

4.4.2 DC Output Characteristics

Table7: DC output characteristics

DC output voltage	20 ~ 35 Vdc
Output voltage ripples	<2.5 V peak-peak at max. load
DC output constant-current	1450 mA, $\pm 5\%$
Output current ripples	<45%
Maximum output power	50 W
Line regulation	$\pm 5\%$
Load regulation	$\pm 5\%$
Start-up time	< 600 ms at 220Vac full load



4.4.3 Protection Specifications

Table8: Protection specifications

AC input overvoltage	275Vac ~ 300Vac
AC input foldback voltage	85Vac ~ 95Vac
DC output open circuit overvoltage	41 ~ 45 Vdc
Thermal foldback temperature	100°C
Lightning surge	Up to 10 kV
Controller IC VDD undervoltage protection	9 Vdc
Controller IC VDD overvoltage protection	24 Vdc

4.4.4 Mechanical Specification

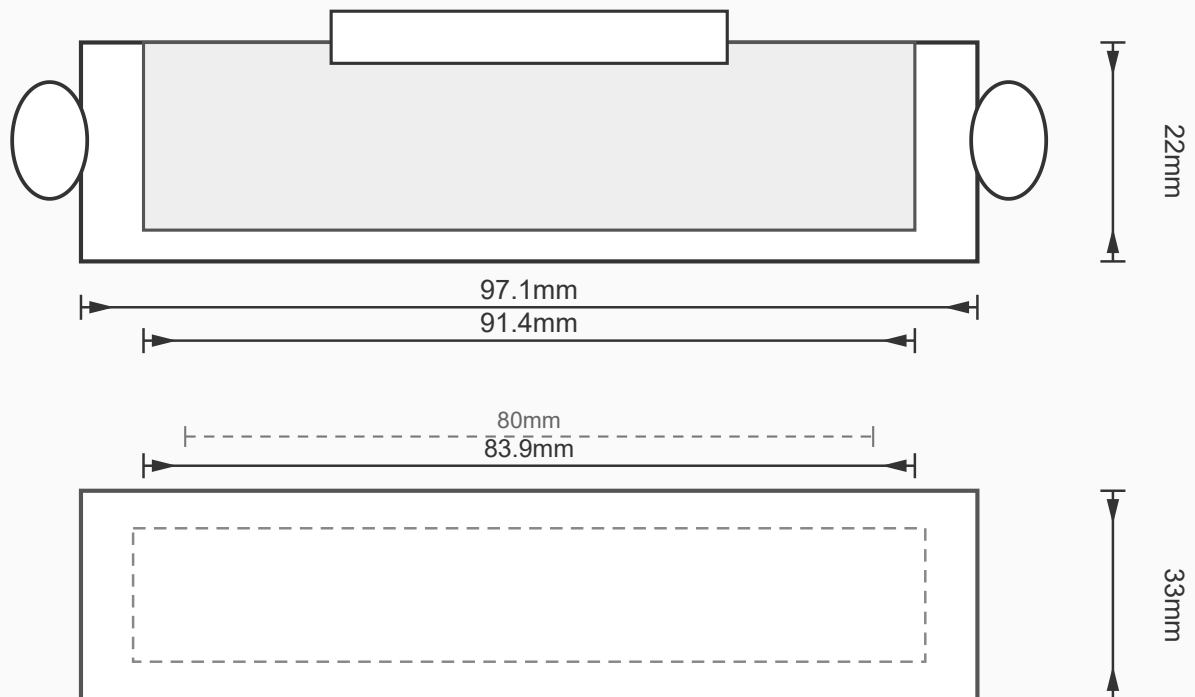


Figure 7: Driver mechanical dimensions in water-proof case IP65